

Faraaz Shaikh

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EDUCATION

M.Sc. Physics

Nowrosjee Wadia College, Pune

Expected July 2027

SGPA: 9.4/10.0 (Semester 2)

Key Coursework: Quantum Mechanics, Mathematical Methods in Physics , Electrodynamics, Classical Physics, Python programming

B.Sc. Physics

Poona College of Arts, Science and Commerce, Pune

Graduated 2025

CGPA: 9.54/10.0

Key Coursework: Nuclear Physics, Mathematical Methods in Physics, Quantum Mechanics, Electrodynamics. Solid state physics, Electronics , Instrumentation.

Particle Physics & Standard Model

NPTEL

Instructor: Dr. Nirmal Raj (IISc)

January 2026 - 24th April 2026

Marks : Awaiting Result

Particle Physics

NPTEL

Instructor: Dr. Mohammad Sajjad Athar (Aligarh Muslim University)

July 2025 - August 2025

Marks: 60%

TECHNICAL SKILLS

Research Tools: Madgraph5, Pythia, Root, TALYS Nuclear Reaction Code , EXFOR Nuclear Data Library

Programming: C, C++, Python (gwosc , pycbc, gwpy, NumPy, Matplotlib, Pandas, Scipy, Qiskit), Linux Bash Scripting, Arduino , LaTeX.

Operating Systems : Linux (Ubuntu 24.04 LTS) , Windows

Industrial Communication Protocols : HART, Modbus.

RESEARCH EXPERIENCE

B.Sc. Research Project

Investigation of Cross Sections of (n, α) and (n, p) on Cd and Hf isotopes using Talys

Computed (n, α) and (n, p) reaction cross sections for Hafnium and Cadmium isotopes using the TALYS nuclear code for application in Generation 4 and Small Modular Reactors.

Analyzed the theoretical validity of the Optical Model, Compound Nucleus, Level density and other reaction models available in Talys by comparing simulation outputs with experimental data from the EXFOR database.

CERTIFICATIONS

Gravitational Wave Open Data Workshop 2026

Completed the data challenge covering gravitational wave parameter estimation, gravitational waveform generation, matched filtering and detector noise characterisation.

Attended Gravitational Wave Open Data Workshop study hub at inter university center of astronomy and astrophysics.

Particle Physics & Standard Model -NPTEL (Prof. Nirmal Raj) - Result Awaited

Particle Physics - NPTEL (Prof. Mohammad Sajjad Athar)

Marks: 60%

Remote Experience for Young Engineers and Scientists

Nuclear Mentoring Program

Transducers for Instrumentation - NPTEL

Marks: 71%

Control Engineering - NPTEL

Marks: 61%

WORK EXPERIENCE

Intern

Krohne Marshall, Pimpri

22nd December 2025 - 8th January 2026

Project: Communication Test System for HART, MODBUS and Fieldbus.

ACADEMIC ACTIVITIES And PRIZES:

Indian Network of Physics Students

Individual Member

Authored "Beyond Standard Model with Neutrino," published in BHOUTIKI Pradnya (College Physics Magazine), discussing the role of seesaw neutrino mass model in CP violation.

2nd Prize in Physiquet 2026 at Nowrosjee Wadia College on **demonstration of BB84 quantum key distribution** as a chat application between 2 computers using IBM's Qiskit python module.